Cell Counting in a Growing Bacterial Colony

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Cell Tracking Undergraduate Challenge

40	640 x 240	3 to 19

Objective

The input is an animated microscopy recording of a growing bacterial colony. Each full frame contains real microscopy on the left and an existing laboratory visualization on the right. This project analyzes only the left microscopy half and automatically estimates the number of visible bacteria in every frame.

Data

The GIF contains 40 frames. Each full frame is 640 x 240 pixels. Cropping the left half produces a 320 x 240 microscopy image. All decoded frames have consistent dimensions. The right-side visualization is not used directly or indirectly in the count.

Approach

A classical computer vision pipeline was selected because the animation is small, no labeled training set is available, and the dark rod-shaped cells can be isolated using local image contrast. The method is deterministic and applies one global parameter set to every frame.

Method

GIF frame → left-side crop → grayscale → Gaussian denoising → local background subtraction → Otsu thresholding → morphological opening → connected components → area filtering → bacteria count

Preprocessing and segmentation

The crop is converted to grayscale. A small Gaussian blur suppresses pixel-scale texture, while a wider Gaussian blur estimates slowly varying illumination. Subtracting the denoised image from this background estimate produces a positive response for locally dark cells. Otsu's method then selects a threshold independently for each frame.

Morphology and counting

A 3 x 3 morphological opening removes isolated texture and severs some narrow foreground bridges. This can help separate touching dark cores, but it can also over-split weak or thin objects. The remaining foreground is labeled with 8-connectivity. Components with areas between 20 and 500 pixels are retained, and their number is reported as the frame count.

Parameters

Parameter	Value	Role
Denoise radius	1 px	Suppress pixel-scale image texture
Background radius	10 px	Estimate slow illumination variation
Threshold	Otsu; floor 12	Select foreground automatically per frame
Morphology	3 x 3 opening	Remove noise and narrow connections
Component area	20-500 px	Reject tiny and overly broad regions
Connectivity	8	Group diagonal foreground pixels

Implementation notes

- The same parameters are applied to all frames; there are no frame-specific corrections.
- Connected-component labeling and Otsu threshold selection are implemented directly in Python and NumPy.
- No pretrained detector, external vision API, watershed, or temporal tracker is used.
- The laboratory's published 2D tracking method was not used to produce this solution.

Results

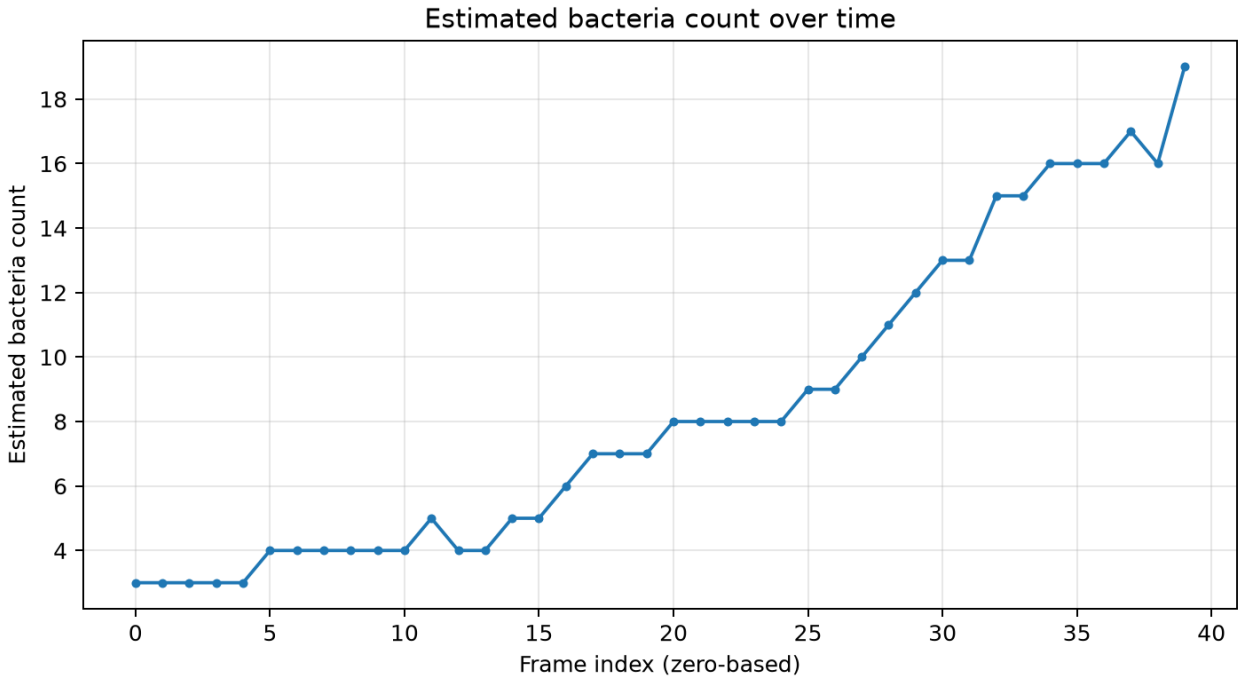


Figure 1. Estimated bacteria count for each zero-based frame index. The count generally increases as the colony grows; monotonic behavior was not imposed.

Frame	Estimated count	Observation
0	3	Small initial colony
20	8	Representative middle frame
39	19	Dense final colony; greatest contact risk
All frames	3-19	40 integers written to counts.txt

Output format

The final counts.txt contains exactly one nonnegative integer per line, with no header, frame identifier, or explanatory text. Its 40 lines match the 40 GIF frames.

Interpretation

Most adjacent-frame changes are zero or +1. The largest increase occurs from frame 38 to 39 (+3), and two transitions decrease by one. These local changes are retained rather than manually smoothed because they expose segmentation uncertainty. Without independent ground truth, the results are treated as plausible estimates rather than an accuracy measurement.

Qualitative Segmentation

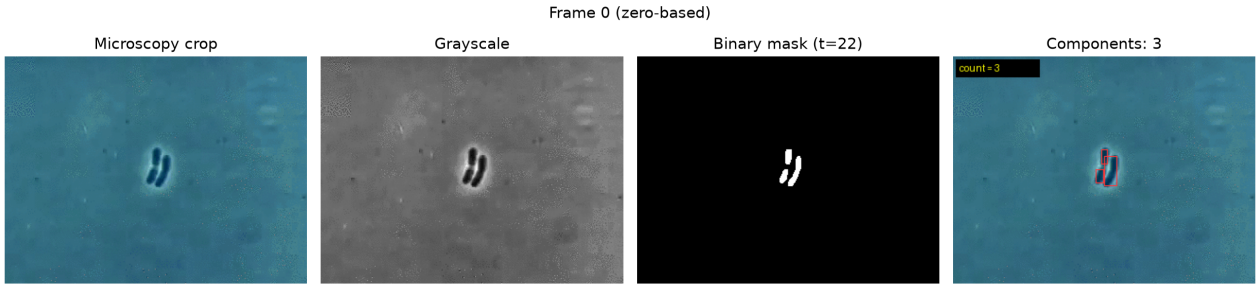


Figure 2. First frame: crop, grayscale, mask, and detection overlay.

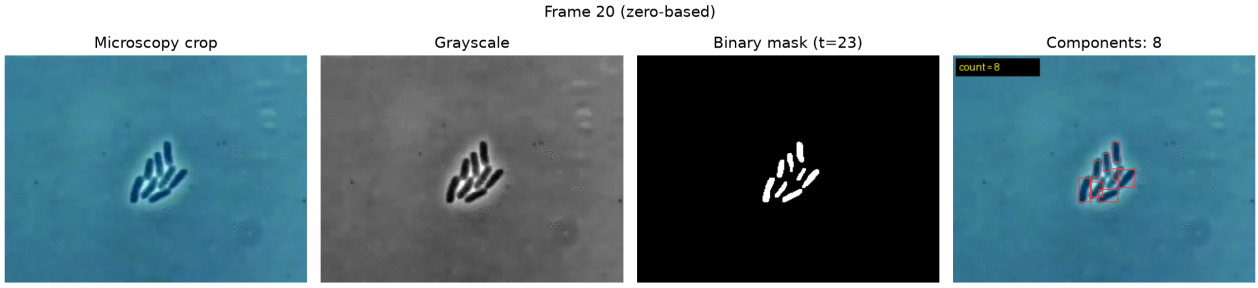


Figure 3. Middle frame: separated dark cores after segmentation.

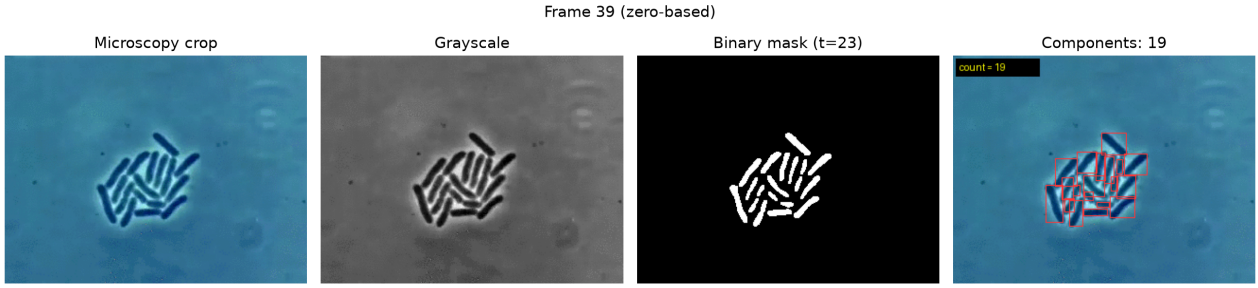


Figure 4. Final frame: a denser colony with more touching structures.

Visual check

The retained components generally align with the dark rod-shaped cells, and background texture is largely excluded. Confidence is lower in the final dense frames, where contact between cells changes the topology of the binary mask.

Failure Analysis and Limitations

Touching cells and dense colonies

Two touching bacteria can form one connected component and cause undercounting. This is the main limitation in later frames. A large merged component may also approach the maximum-area cutoff.

Fragmentation and overcounting

Opening can correctly break a narrow bridge between cells, but it can also split a weak or thin response. Faint components near the minimum-area threshold may appear in one frame and disappear in the next.

Contrast and generalization

Local background subtraction stabilizes uneven illumination, but a low-contrast cell can still fall below the threshold. The area and scale parameters were selected for this GIF and may not transfer unchanged to another microscope, magnification, or cell morphology.

Ground truth

No annotated counts were provided. Therefore, this report does not claim an accuracy percentage or optimal performance.

Possible Improvements

- Annotate representative early, middle, and dense late frames for quantitative evaluation.
- Test marker-controlled watershed only on confirmed merged components.
- Use temporal consistency to distinguish cell division from segmentation flicker.
- With sufficient labeled data, compare against learned instance segmentation or tracking.

References

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4. Otsu, N. (1979). A Threshold Selection Method from Gray-Level Histograms. IEEE Transactions on Systems, Man, and Cybernetics, 9(1), 62-66. https://doi.org/10.1109/TSMC.1979.4310076
5. Matplotlib. savefig. https://matplotlib.org/stable/api/_as_gen/matplotlib.pyplot.savefig.html